RDF4RiskAssessment Toolkit: A toolkit for converting tabular research data to FAIR RDF for risk assessment and life sciences

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Abstract

Risk assessment is crucial for consumer health protection and food safety within the Risk Analysis framework. It requires integrating heterogeneous data from multiple scientific disciplines. However, research data often exists in non-standardized formats, creating data silos that hinder cross-institutional reuse and AI-based analysis. We developed the RDF4RiskAssessment Toolkit within the KIDA project to transform tabular research data into FAIR-compliant Linked Data and used it in the AMBROSIA project to harmonize concepts. The toolkit features a four-stage workflow: matching table generation, term reconciliation via external ontology services, configurable RDF generation, and bidirectional RDF-table conversion. It supports metadata standards (DCAT, Dublin Core, BIBO) and vocabulary alignment (SKOS). Testing with 14 datasets from 6 publications yielded approximately 1,860 mapped vocabulary concepts. The toolkit enables researchers without Semantic Web expertise to produce machine-readable, interoperable datasets for risk assessment and life sciences.

Keywords

Linked Open Data, Data harmonization, Consumer health protection, Ontology Mapping, Semantic interoperability

1. Introduction

Risk assessment is central to consumer protection and food safety within the Risk Analysis framework [1]. It increasingly depends on the integration of data from chemistry, life sciences, environmental sciences, and social sciences. Despite the growing availability of data, research often relies on spreadsheets or isolated databases, leading to data silos and reduced interoperability. This fragmentation hinders the use of AI-based methods, collaborative evaluation, and effective decision-making [2]. Yet turning such tabular data into interoperable Linked Data remains difficult due to technical complexity and the lack of accessible tools tailored to small-scale volatile datasets. To address this gap, we developed the RDF4RiskAssessment Toolkit within the KIDA project, a pilot project aimed at accelerating AI and data interoperability adoption among German Federal research institutes [3]. Additionally, the toolkit was used to harmonize concepts from agriculture and predictive microbiology in the AMBROSIA Project [4]. The toolkit converts tabular data into machine-readable Linked Data through a user-friendly software that does not require prior Semantic Web expertise.

2. Tool Description

The toolkit provides web-based services that support a four-stage semantic transformation process (Figure 1). First, the "Matching Table Generation" service ingests CSV/Excel files and produces a cleaned dataset, along with a candidate-term table for ontology matching derived from column headers and

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unique cell values. This table follows core concepts of the SSSOM mapping model, but omits optional metadata to keep curation lightweight.

Second, the candidate-term table is used in the "Reconciliation Service" to link dataset terms to authoritative ontologies via API endpoints (Wikidata, BioPortal, AgroPortal, EarthPortal, NCBI). Users select endpoints and candidate matches in a semi-automatic interface that supports string similarity scoring (e.g., Levenshtein). Selected matches are annotated with semantic relations and stored back into the table for use in the RDF generation.

Third, the original data and the reconciled candidate-term table are combined by the "RDF Generation Service" into a row-wise RDF representation: each row becomes a subject URI, columns become predicates, and cells become objects. This allows a fast and automatic conversion without forcing users to readjust the original data. The matchings of the candidate-table are materialized as SKOS concepts and linked to the original data to preserve and harmonize the mappings. Additionally, DCAT metadata is generated for the RDF dataset describing general information. The generated output can be downloaded in a common RDF serialization such as TriG, Turtle, or JSON-LD. For advanced cases, an optional custom mapping schema (configuration rules comparable to declarative mappers such as YARRRML [5]) supports blank nodes, nested structures, and subject URIs derived from existing identifiers or shared keys.

Finally, the "RDF-to-Table Conversion service" supports round-trip validation and accessibility by reconstructing RDF back into a tabular form for human review, without requiring SPARQL.

By enabling semantic enrichment and improving cross-disciplinary data interoperability, the RDF4RiskAssessment toolkit contributes to more accessible, reusable, and integrated scientific datasets for risk assessment and life sciences.

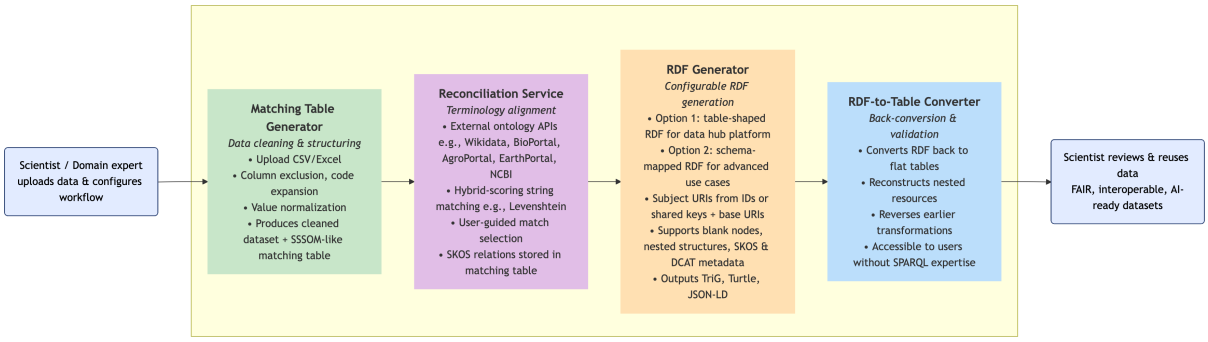

Figure 1: This flowchart illustrates the modular architecture of the RDF4RiskAssessment Toolkit, designed to facilitate the transformation of scientific datasets into RDF.

3. Results

Throughout the KIDA project, 14 datasets from 6 publications were converted into Linked Data, yielding approximately 1,860 vocabulary concepts mapped against Wikidata and OBO ontologies. The largest data graph contained 140,699 triples; the smallest contained 189. Data spanned diverse domains from wine authenticity to epidemiological analyses, demonstrating the toolkit's adaptability across domains and scales. . The resulting RDF supports SPARQL queries, cross-disciplinary data exchange, and linking across datasets via shared URIs from reconciliation. In AMBROSIA, an EU project, the initial SKOS vocabularies were reconciled against external ontologies using the Reconciliation Service, enabling ongoing term enrichment and mapping (e.g., Wikidata, OBO). The source code of the project is available at: https://github.com/KIDA-BfR/RDF4Risk-Toolkit

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Declaration on Generative AI

During the preparation of this work, the authors used a Llama-3.3-70B Model provided within the KIDA project in order to check grammar, improve fluency and spelling of the text. The authors reviewed and edited the content as needed and takes full responsibility for the publication's content.

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